

Enabling Branching Support in Server-Side Vitruvius

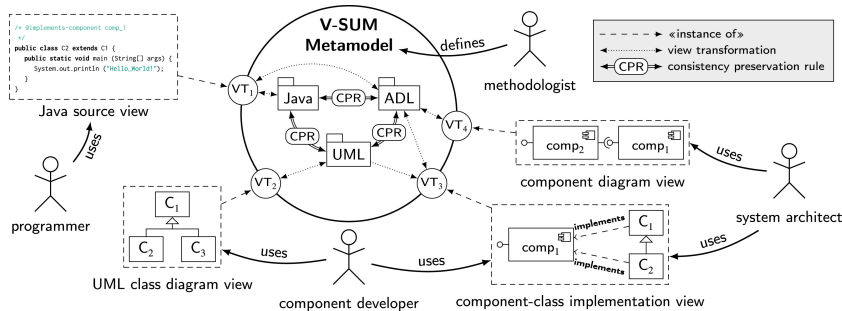
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Motivation

- Vitruvius for consistency preservation
- No native branching or versioning on Server-side
- Git cannot ensure consistency on its own



Source: Enabling consistency in view-based system development - The Vitruvius approach

- **Problem:** No support for parallel change propagation
- **Idea:** Git for branching and history, Vitruvius for delta handling and validation
- **Benefit:** Safe parallel work, early conflict detection, replay and rollback
- **Action:** Branch integration, hybrid storage, model-aware merge support

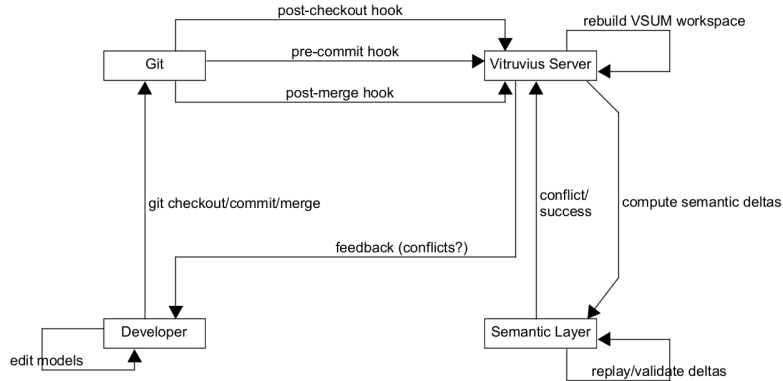
Foundation

- Vitruvius: operation-based change and consistency maintenance (VSUM)
- EMF (Eclipse Modeling Framework) for metamodels
- Git: branch DAG, commit history, diff, merge
- State snapshot vs. change operation

Related Work

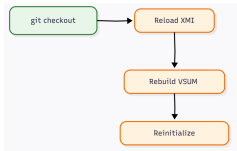
- Git: branching, DAG history, not model-aware
- EMF Compare / DiffMerge: model-aware diffs, no history management
- AMOR / EML: semantic merge patterns, conflict resolution
- CDO: model repository
- Gap: no integrated solution

Example Workflow

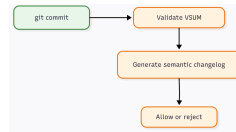


Implementation Concept

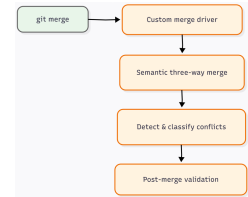
- Git: handles branches, commits, history
- Vitruvius: model-aware change handling
- Git hooks: connect file-level actions with model-level logic:
 - post-checkout: rebuild VSUM workspace
 - pre-commit: validate & extract deltas
 - post-merge: semantic check
- Custom merge driver: model-aware merging



(a) Post-checkout



(b) Pre-commit



(c) Post-merge

Hybrid Storage Model

- Git: XML snapshots + history
- Vitruvius: changelogs, other metadata

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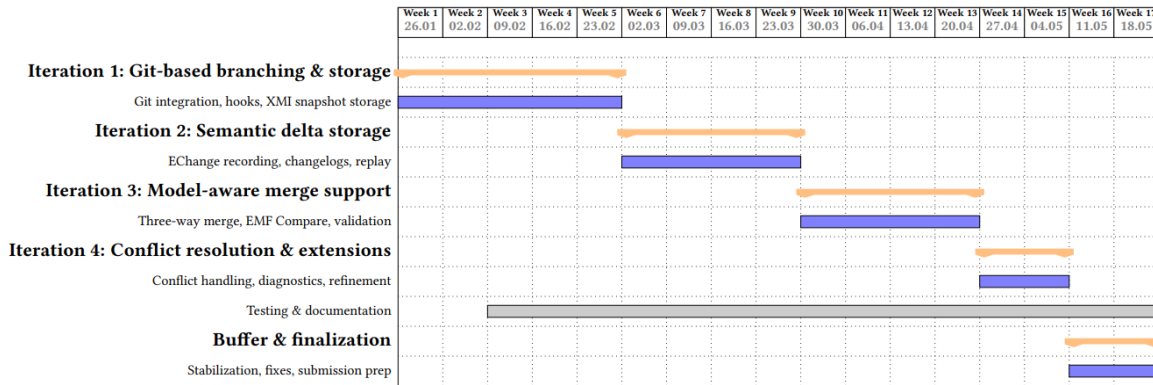
vitruvius-project/
├─ .git/                # git repository
│  └─ hooks/            # integration points
│     │ └─ pre-commit    # consistency validation
│     │ └─ post-checkout # workspace rebuild on branch switch
│     │ └─ post-merge    # semantic merge validation
│     └─
│  └─ refs/              # branch and tag pointers
│  └─ objects/           # versioned model content
│  └─
├─ .gitattributes        # custom merge drivers
├─
├─ .vitruvius/           # vitruvius semantic layer
│  └─ branches/          # branch metadata
│  └─ changelogs/        # operation-based deltas (EChange)
│  └─ merges/            # merge metadata and conflicts
│  └─ versions/          # version tags and snapshots
└─ metamodels/          # versioned EMF metamodels

```

Evaluation

- Branch operations
- Merge scenarios
- VSUM consistency

Timeline



Summary

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